CSE 360 - TEAM PROJECT PHASE 1

Input Validation

Complete TP1 text-input inventory, rules, user feedback, and verification

20	19 + 1	46 / 46	33 / 33
text-input objects	reachable + dormant	validation tests	OTP tests

Prepared by	Vishwam (V.G.)

Completion statement. Every text-input object in `TP1/src` was located and classified. Reachable input is bounded before expensive processing or database use, shared rules are reused across screens, and invalid input produces useful feedback without committing a write. The single dormant inherited dialog is disclosed explicitly so the inventory is object-complete.

Implemented	Dynamic password help now covers all four new-password workflows: First Admin, invited account, User Update, and OTP reset.

1 - Validation design and shared rules

Validation follows one consistent order: **(1)** null/blank and reasonable length checks, **(2)** syntax or finite-state-machine checks, **(3)** cross-field or database constraints, and only then **(4)** the requested operation. Existing login credentials deliberately receive only bounds and authentication checks; rejecting them for current creation rules could lock out a legitimate older account.

Shared validator	Accepted input / boundary	Where enforced before use
UserName FSM	4–32 characters; starts with ASCII letter; later characters are letters, digits, or **-** _ **.** **&**; a separator must be followed by alphanumeric.	First Admin, invited account, and username update. Detailed error includes the one-based failure position.
New password	8–20 characters; at least one uppercase, lowercase, digit, and permitted special character. Null, empty, invalid-character, 21-character, and 10,000-character inputs are rejected.	Shared dynamic popup on First Admin, invited account, User Update, and OTP reset; submit-time recheck remains defense-in-depth.
Email recognizer	254 total / 64 local-part maximum; one @; structured local part and domain; no leading, trailing, or doubled period; domain contains a period; TLD has at least two letters.	Admin invitation and email update. Duplicate outstanding invitation is separately refused.
Personal name	Optional in the present TP1 implementation; if supplied: at most 50 characters and only ASCII letters, spaces, hyphens, and apostrophes.	First, middle, last, and preferred-first-name update dialogs before any database write.
Deadline	Noneditable date plus trimmed 24-hour **HH:mm** text (maximum 5 characters); resulting time must be in the future and no more than 30 days away.	Invitation and one-time-password deadline screens share the same parser and policy.
Login lookup	Existing-credential lookup uses username ≤32 and password/OTP ≤64. Composition is intentionally not revalidated at login.	Login controller before database use; all failures use one generic message to avoid revealing whether an account exists.
Invitation code	Non-null, nonblank, 1–10 characters; then checked against the live invitation table and deadline.	Login account-setup action before the New Account screen opens.

Dynamic password behavior

The shared popup masks characters and reevaluates on every edit. Six visible requirements start unsatisfied; Finish remains disabled until all are true. Every opening resets the field, messages, flags, and button. Null, empty, or excessive input clears earlier green state. Error positioning uses bullets rather than echoing secret text, and plaintext is not written to the console. First Admin and invited-account screens retain a separate masked confirmation field with its own 20-character bound and a submit-time evaluator check.

One-time password boundary

A generated OTP is 12 characters and follows the same password composition policy. At login, the database uses one conditional update to both validate and clear it. Therefore only one simultaneous request can win. Cancellation cannot restore the OTP; a successful reset returns to a blank login screen and requires the new permanent password.

2 - Complete text-input inventory

The table below accounts for every `TextField`, `PasswordField`, and `TextInputDialog` created by production code. Field 20 is intentionally included even though it is unreachable, because the active password-update button uses the shared dynamic popup.

#	Screen / field	Control	Validation before use	Failure / success behavior	Code path
1	Login · Username	TextField	≤32 before lookup; unknown, malformed, empty, or long input is refused.	Generic credential alert; no account-enumeration detail.	ControllerUserLogin.doLogin
2	Login · Password / OTP	PasswordField	≤64 before lookup. Active OTP is tested/consumed first; otherwise permanent password is authenticated.	Generic credential alert. OTP success forces reset, then fresh login.	ControllerUserLogin; Database
3	Login · Invitation code	TextField	Nonblank, 1–10; then live-code and deadline checks.	Specific boundary, unknown-code, or expired-code alert.	validInvitationInput; ViewNewAccount
4	First Admin · UserName	TextField	Full 4–32 UserName FSM.	Detailed reason + error position; no account is written.	ControllerFirstAdmin
5	First Admin · Chosen password	Read-only PasswordField	Value can only come from the live 8–20 password popup; submit-time evaluator rechecks it.	Masked value; live red/green rules; setup blocked on cancel/invalid.	View/ControllerFirstAdmin
6	First Admin · Confirm password	PasswordField	Explicit ≤20 bound, then exact equality with field 5.	Overlong confirmation is cleared with a size message; mismatch clears both fields.	ControllerFirstAdmin
7	Invited Account · UserName	TextField	Full UserName FSM plus duplicate database check.	Detailed error; duplicate no longer terminates the app.	ControllerNewAccount
8	Invited Account · Chosen password	Read-only PasswordField	Value can only come from live popup; submit-time evaluator rechecks it.	Masked value; live rules; setup blocked on cancel/invalid.	View/ControllerNewAccount
9	Invited Account · Confirm password	PasswordField	Explicit ≤20 bound, then exact equality with field 8.	Overlong confirmation is cleared with a size alert; mismatch clears both fields.	ControllerNewAccount
10	Admin Home · Invitation email	TextField	Full shared email recognizer; duplicate outstanding email check.	First structural error is explained; invitation is not created.	ControllerAdminHome
11	Admin Home · Invitation time	TextField	≤5, trimmed HH:mm; future and ≤30 days with selected date.	Malformed, past, and too-far deadlines receive distinct messages.	ControllerAdminHome
12	OTP Page · Deadline time	TextField	Same shared parser and 30-day policy as invitation time.	OTP is not stored; error is shown on the OTP page.	ControllerOneTimePassword
13	Dynamic Popup · New password	PasswordField	Every edit applies 8–20 composition rules; excessive/null/empty input clears all prior validity state.	Masked; red/green rules; Finish disabled until valid; reopening starts clean.	passwordPopUpWindowModel/View
14	User Update · UserName	TextInputDialog	Full shared UserName FSM plus duplicate check.	Specific invalid/duplicate alert; no database write.	ControllerUserUpdate; ViewUserUpdate
15	User Update · First name	TextInputDialog	Optional; if supplied ≤50 and allowed name characters only.	Invalid entry produces a Name alert; value is unchanged.	ControllerUserUpdate.checkName
16	User Update · Middle name	TextInputDialog	Optional; if supplied ≤50 and allowed name characters only.	Invalid entry produces a Name alert; value is unchanged.	ControllerUserUpdate.checkName
17	User Update · Last name	TextInputDialog	Current TP1 behavior: optional; if supplied ≤50 and allowed name characters only.	Invalid entry produces a Name alert; value is unchanged.	ControllerUserUpdate.checkName
18	User Update · Preferred first name	TextInputDialog	Optional; if supplied ≤50 and allowed name characters only.	Invalid entry produces a Name alert; value is unchanged.	ControllerUserUpdate.checkName
19	User Update · Email	TextInputDialog	Full shared email recognizer.	Specific invalid-email alert; value is unchanged.	ControllerUserUpdate.checkEmailAddress
20	User Update · Legacy password dialog	Dormant TextInputDialog	Instantiated by inherited code but never connected to a button or shown.	No user can reach it. The active Update Password action uses field 13 instead.	ViewUserUpdate (dead control)

3 - Non-text controls and remaining checks

Not every control needs a text recognizer. The two deadline DatePickers are explicitly noneditable; users select a date. User, role, deletion target, and multi-role ComboBoxes remain noneditable and only return team-provided choices. The OTP target must be a real selected user, and invitation roles come from Admin, Contributor, Viewer, and Curator. These controls still receive null/placeholder and authorization checks before use.

Control family	Why arbitrary text is impossible	Remaining checks
Deadline DatePicker	`setEditable(false)`; calendar selection only.	Date present; combined deadline future and ≤30 days.
Role / user ComboBoxes	Default noneditable JavaFX choice controls populated by application data.	Selection non-null; placeholder refused; database authorization/policy applied.
Confirmation Alerts	Fixed Yes/No or OK choices rather than typed text.	Destructive operation proceeds only after explicit Yes.

Reasonable-size defense	All reachable typed values are bounded before database use. Validators short-circuit excessive text, including 10,000-character attack cases.

4 - Verification evidence

Evidence	Scope	Result
InputValidationTestingAutomation 1–46	UserName, password, email, names, update reuse, null/invalid password input, chosen/confirmation 20/21 boundaries, 10,000-character attacks, and stale dynamic-state regression.	46 passed 0 failed
OneTimePasswordTestingAutomation 1–33	Generation, future/expired deadlines, wrong/null/oversized input, immediate one-use consumption, permanent-password reset, replacement, and two concurrent redemption attempts.	33 passed 0 failed
LoginInputTests 1–7	Invitation-code null, empty, blank, valid lengths 1/6/10, and rejected length 11.	7 passed 0 failed
TP1RegressionSuite	All six team automation classes after integration.	152 passed 0 failed
DynamicPasswordViewTests P01–P06	Masked password control, valid-to-empty and overlong transitions, clean reopen state, and redacted invalid-input feedback.	6 passed 0 failed
FirstUserReloginTests R01–R04	First-user completion label, return to login, cleared authorization, and singleton reset for ordinary account updates.	4 passed 0 failed

5 - Code traceability

Concern	Primary implementation	Executable evidence
Identity + passwords	UserNameRecognizer; password popup Model/View; FirstAdmin, NewAccount, Login, and UserUpdate controllers	InputValidation 1–11, 35, 37–46; Dynamic UI P01–P06
Profile + invitations	EmailAddressRecognizer; ControllerUserUpdate.checkName; AdminHome deadline rules; Login invitation boundary	InputValidation 12–34, 36; LoginInput 1–7
OTP + secure completion	Database; Login and OneTimePassword controllers; FirstAdmin and UserUpdate completion flow	OneTimePassword 1–33; FirstUserRelogin R01–R04